

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-I	20.00	31.58
2	Module-III	20.00	26.32
3	Module-II	20.00	21.05
4	Module-IV	20.00	13.16
5	Module-V	20.00	7.89

Weightage representation okay

: Weightage representation exceeded more than 5%

:Weightage representation exceeded more than 10%

CSE601

[ET]

Enroll. No. -

END SEMESTER EXAMINATION : April-May, 2026

Data Structures and Algorithm Design

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.

(a)

What is asymptotic notation?

(b)

Write the time complexities of all operations in a singly linked list.

(c)

What is divide and conquer?

(2)

Delete

(b)

Write the time complexities of all operations in a singly linked list.

(2)

Delete

(c)

What is divide and conquer?

(2)

Delete

Q2.

(a)

Convert Postfix expression of the expression (a+b) * (c+d)/f and A+B*(C-D)/(P-R)

(b)

Define AVL Tree.

(4)

Delete

(b)

Define AVL Tree.

(2)

Delete

Q3.

(a)

Define the Pivot element in the Quick Sort Algorithm.

(b)

Define Dynamic Programming.

(c)

Define the 0/1 Knapsack Problem.

(2)

Delete

(b)

Define Dynamic Programming.

(2)

Delete

(c)

Define the 0/1 Knapsack Problem.

(2)

Delete

Q4.

(a)

Write a recursive method to find the factorial of a given number.

(b)

Define String Matching Algorithm and Approximation Algorithm.

(3)

Delete

(b)

Define String Matching Algorithm and Approximation Algorithm.

(3)

Delete

Q5.

Define NP, NP-Complete and NP-Hard problems with examples.

(6)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.

(a)

Write In-Order, Pre-Order and Post-Order Binary Tree Traversal Algorithms.

(b)

Discuss the different rotations in the AVL Tree with an example. Write code snippets for each rotation.

(5)

Delete

(b)

Discuss the different rotations in the AVL Tree with an example. Write code snippets for each rotation.

(5)

Delete

Q7.

(a)

Write a complete Java Program for Stack operations.

(b)

Discuss the Heap sort algorithm analysis with example data.

(5)

Delete

(b)

Discuss the Heap sort algorithm analysis with example data.

(5)

Delete

Q8.

(a)

Write Java code to find the factorial of a given number using Dynamic Programming.

(b)

Define the Shortest Path Algorithm and write Dijkstra’s Algorithm to find the shortest path from a single source vertex to all other vertices.

(5)

Delete

(b)

Define the Shortest Path Algorithm and write Dijkstra’s Algorithm to find the shortest path from a single source vertex to all other vertices.

(5)

Delete

Section - C : Compulsory question

Q9.

(a)

Develop Java Code to delete the last node of the given Singly Linked List head.

(b)

Develop Java code for Enqueue() and Dequeue() operations

(c)

Define recursive functions and their types with examples.

(d)

Discuss Time Complexity analysis for different search algorithms.

(4)

Delete

(b)

Develop Java code for Enqueue() and Dequeue() operations

(4)

Delete

(c)

Define recursive functions and their types with examples.

(4)

Delete

(d)

Discuss Time Complexity analysis for different search algorithms.

(4)

Delete